

STUDENT ACTIVITY HANDOUT

IFY STATISTICS | SEMESTER 1 - WEEK 1A

Mathematical Foundations I: Sets, Counting and Arithmetic

Bellerbys
Preparation for University Education

Name _____ Class _____ Date _____

EXERCISE 1

Symbol Decoder

Write each statement as a complete English sentence. Then compare your sentences with a partner.

1. $4 \in A$ YOUR SENTENCE _____

2. $7 \notin A$ YOUR SENTENCE _____

3. $n(A) = 5$ YOUR SENTENCE _____

4. $B = \emptyset$ YOUR SENTENCE _____

LANGUAGE HELP $\in$ is an element of $\notin$ is not an element of
 $n(A)$ number of elements in A $\emptyset$ empty set

SETS AND VENN DIAGRAMS

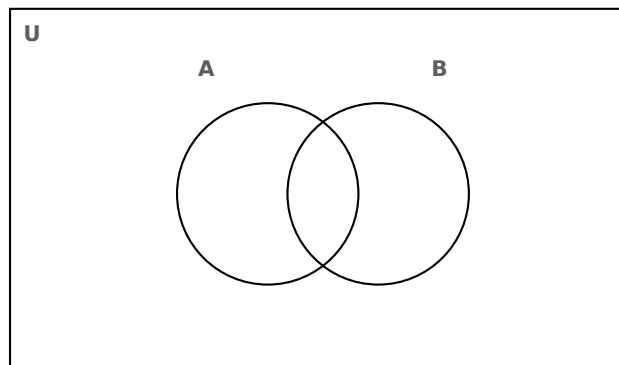
EXERCISE 2

Venn Detective

Use the defined universal set. Write each set in braces and do not repeat elements.

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 2, 4, 6, 8\} \quad B = \{2, 3, 6, 7, 8\}$$



1. Find $A \cap B$

2. Find $A \cup B$

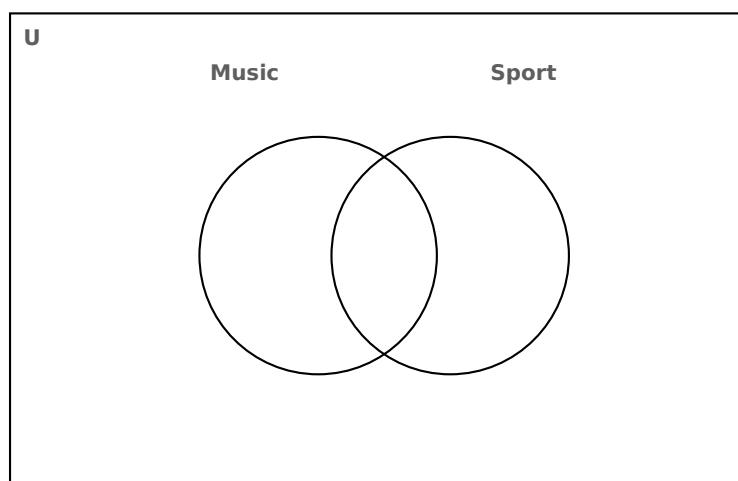
3. Find A^c

4. Find $n(A \cup B)$

EXERCISE 3

Class Survey Sets

A class contains 30 students. $n(M) = 18$ join Music; $n(S) = 14$ join Sport; $n(M \cap S) = 6$ join both groups.



1. Draw and label a Venn diagram.

2. Find Music only and Sport only.

3. Find $n(M \cup S)$.

4. Find how many join neither group.

CHECK YOUR TOTAL

Music only + both + Sport only + neither should equal the class total.

$$\underline{\quad\quad} + \underline{\quad\quad} + \underline{\quad\quad} + \underline{\quad\quad} = 30$$

BASIC COUNTING AND FACTORIALS

COUNTING METHOD	WHEN TO USE IT
Basic counting	Separate stages: multiply the number of choices at each stage.
Factorial	Arrange every available item: $n! = n \times (n-1) \times \dots \times 1$.
Permutation	Order or different roles matter: swapping people creates a new result.
Combination	Equal-role group: swapping selected people does not create a new group.

EXERCISE 4Basic Counting Practice

For each situation, write the number of choices at each stage. Then multiply them.

1. Choose 1 snack from 4 and 1 drink from 3.

CHOICES

TOTAL

2. Choose one digit 0-9, then one letter A-F.

CHOICES

TOTAL

3. Create a 3-digit PIN; 0 may be first and repeats are allowed.

CHOICES

TOTAL

4. Challenge: two letters followed by three digits; repeats allowed.

CHOICES

TOTAL

EXERCISE 5Factorial Practice

Write the multiplication before the factorial answer.

1. Arrange 4 different books in a row.

EXPANSION AND ANSWER

2. Arrange 6 students in a line.

EXPANSION AND ANSWER

3. Arrange 8 different photographs in a row.

EXPANSION AND ANSWER

4. Challenge: 6 books are arranged, but Statistics must be first.

EXPANSION AND ANSWER

IFY Statistics | Mathematical Foundations I

Student Activity Handout 3

PERMUTATIONS AND COMBINATIONS

EXERCISE 6

Permutation Practice

Use a permutation when order or different roles matter. State what changes when two selected items are swapped.

1. Choose a president and vice-president from 8 students.

METHOD, WORKING AND ANSWER

2. Arrange 4 letters selected from 9 different letters.

METHOD, WORKING AND ANSWER

3. Award gold, silver and bronze to 6 teams.

METHOD, WORKING AND ANSWER

4. Challenge: how many 5-digit numbers have no repeated digits?

The first digit cannot be 0.

METHOD, WORKING AND ANSWER

EXERCISE 7

Combination Practice

Explain why order does not matter before calculating. Use a combination for equal-role groups.

1. Choose 3 students from 8 for one team.

WHY ORDER DOES NOT MATTER

CALCULATION

2. Choose 4 survey participants from 10 students.

WHY ORDER DOES NOT MATTER

CALCULATION

3. Choose 2 different toppings from 7.

WHY ORDER DOES NOT MATTER

CALCULATION

4. Challenge: choose 2 of 5 Year-1 and 2 of 6 Year-2 students.

WHY ORDER DOES NOT MATTER

CALCULATION

CHOOSING A COUNTING METHOD

EXERCISE 8

Choose, Explain, Calculate

For each problem, tick one or more methods before calculating. Write one sentence explaining why each selected method fits.

1. Choose 1 main from 5, 1 drink from 4 and 1 dessert from 2.

METHOD ☐ Basic counting ☐ Factorial ☐ Permutation ☐ Combination

WHY _____

CALCULATION _____

2. Choose captain and vice-captain from 9 students.

METHOD ☐ Basic counting ☐ Factorial ☐ Permutation ☐ Combination

WHY _____

CALCULATION _____

3. Choose 2 equal-role delegates from 9 students.

METHOD ☐ Basic counting ☐ Factorial ☐ Permutation ☐ Combination

WHY _____

CALCULATION _____

4. Arrange all 7 laboratory samples in a row.

METHOD ☐ Basic counting ☐ Factorial ☐ Permutation ☐ Combination

WHY _____

CALCULATION _____

5. Challenge: choose a 3-person committee from 8, then its leader.

METHOD ☐ Basic counting ☐ Factorial ☐ Permutation ☐ Combination

WHY _____

CALCULATION _____

LEADER CHECK Do you need one method or more than one? Think about choosing the committee first, then choosing its leader. Explain each decision before calculating.

TYPES OF NUMBERS

Number Types: One Reference Table

Type	Definition	Examples	Important detail
Natural	counting numbers	1, 2, 3, ...	positive whole numbers; this course starts at 1
Whole	natural numbers and 0	0, 1, 2, ...	non-negative integers
Integer	negative, zero, positive	−5, 0, 12	no fractional part
Rational	can be written p/q	3/4, −2, 0.25	p, q integers; q ≠ 0; decimal terminates or repeats
Irrational	cannot be written p/q	$\sqrt{2}$, π	decimal is non-terminating and non-repeating
Real	rational and irrational	all examples above	all points on the number line

NESTED GROUPS Natural $\subset$ Whole $\subset$ Integer $\subset$ Rational $\subset$ Real
−2 = −2/1 and 0.25 = 1/4, so both are rational.

EXERCISE 9Simplify, Then Classify

Simplify each value first. Then write the smallest category that contains the answer.

Original value	Simplified value	Smallest category
1. 0		
2. $-\sqrt{64}$		
3. $\sqrt{(49/4)}$		
4. 0.272727... (27 repeats)		
5. $\sqrt{18}$		
6. π		

PEMDAS PRACTICE

PEMDAS REFERENCE

Parentheses | Exponents | Multiplication and Division | Addition and Subtraction
 Multiplication and division have equal priority; addition and subtraction do too. Work left to right.
 Please Excuse My Dear Aunt Sally

EXERCISE 10 BODMAS / PEMDAS Error Clinic

Find the first incorrect step, write the corrected calculation and explain the rule.

1. $3 + 5 \times 2 = 8 \times 2 = 16$

FIRST ERROR _____ CORRECT ANSWER _____

EXPLANATION _____

2. $24 \div 6 \times 2 = 24 \div 12 = 2$

FIRST ERROR _____ CORRECT ANSWER _____

EXPLANATION _____

3. $18 - 6 + 2 = 18 - 8 = 10$

FIRST ERROR _____ CORRECT ANSWER _____

EXPLANATION _____

EXERCISE 11 PEMDAS Challenge

No calculator. Show one operation per line. Check brackets first and equal-priority operations left to right.

1. $36 \div 3 \times (2 + 4) - 5^2$

WORKING

2. $5 + 3[8 - 2(3 + 1)]$

WORKING

3. $120 - \{4[3^2 + 6] - 8\} \div 2$

WORKING

FINAL CHALLENGE AND EXIT TICKET

EXERCISE 12

Brain-Boiling Challenge

No calculator. Show every step. Remember: multiplication and division are equal-priority operations, as are addition and subtraction.

$$[(16 - 8) \div 2] + 5 \times 11 - 4^2 - \sqrt{(8 \times 2)}$$

FULL WORKING

EXERCISE 13

Independent Exit Ticket

Work independently. Write the method or main steps, not only the final answer.

1. $A = \{1, 2, 4, 8\}$, $B = \{2, 3, 4, 6\}$. Find $A \cup B$.

METHOD, WORKING AND ANSWER

2. Choose and calculate: select 3 equal-role students from 8.

METHOD, WORKING AND ANSWER

3. Simplify and classify $\sqrt{27}$.

METHOD, WORKING AND ANSWER

4. Calculate $48 \div [2 \times (3 + 1)] + 5^2$.

METHOD, WORKING AND ANSWER
